TDA on CORe50 streams: final results

CORe50 128×128 · Bocconi HPC cluster · 4 November 2026

Setup

- **Data:** CLIP features (RN50, ViT-B/16) of all 11 sessions, extracted once. Results on the 3 test sessions (3, 7, 10): 44,972 frames, 50 objects, 10 classes.
- **Hyper-parameters:** chosen on the 8 non-test sessions (one frame in five, order S0, 5 seeds) among the 10 official TDA configs, then frozen. **caltech101** wins with both backbones (appendix).
- **Statistics:** 50 seeds per setting, mean and 95% confidence interval.
- **Methods:** CLIP zero-shot · TDA · **M2** (cache votes with the per-class mean instead of the sum) · **Oracle** (cache filled with the true labels: upper bound of the cache) · **Smooth** (CLIP averaged over the last N frames, the classic video baseline).
- **Stream orders:** S0 all shuffled (as in the paper) · S1 class persists with probability ρ · S2 one session at a time, shuffled inside · S3 one session at a time, object videos in temporal order.

Main results: accuracy (%) by stream order

Order	RN50 (CLIP 73.37)			ViT-B/16 (CLIP 79.17)		
	TDA	M2	Oracle	TDA	M2	Oracle
S0	79.35 ±0.06	79.35 ±0.06	79.22 ±0.08	84.81 ±0.08	84.81 ±0.08	85.07 ±0.11
S1, ρ = 0.5	79.35 ±0.07	79.35 ±0.07	79.26 ±0.09	84.87 ±0.09	84.87 ±0.09	84.98 ±0.11
S1, ρ = 0.9	79.31 ±0.07	79.31 ±0.07	79.28 ±0.09	84.84 ±0.09	84.84 ±0.09	84.99 ±0.11
S2	**80.18 ±0.11**	**80.19 ±0.11**	80.21 ±0.12	**86.08 ±0.17**	**86.09 ±0.17**	85.82 ±0.14
S3	79.68 ±0.13	79.62 ±0.13	**81.29 ±0.21**	85.49 ±0.22	85.45 ±0.22	**86.29 ±0.21**

CLIP gives the same accuracy for every order (sanity check passed). ± = 95% CI over 50 seeds. Green: the only clear oracle gain.

Video baseline: CLIP smoothed over the last N frames

Window	RN50, S3	RN50, instance	ViT, S3	ViT, instance
5 frames	74.86	74.86	79.73	79.73
15 frames	75.60	75.63	**80.40**	**80.41**
30 frames	**76.56**	**76.61**	80.02	80.05
TDA on S3, for comparison	*79.68*		*85.49*	

CIs ≤ ±0.03. "instance" is the instance order defined in `streams.py`.

What we see

1. **TDA helps a lot on CORe50:** +5.9 to +6.8 points over CLIP with RN50, +5.6 to +6.9 with ViT-B/16.
2. **Order matters a little, and session context helps:** S2 is the best order with both backbones (+0.83 RN50, +1.27 ViT over S0). Temporal order (S3) is in between; synthetic label persistence (S1) has no effect.
3. **M2 changes nothing:** it matches TDA within 0.06 points in every setting. On CORe50 the coverage bias that M2 targets does not seem to occur; per-class cache counts should tell us why.
4. **Cache label noise is not the bottleneck, except in S3:** with true labels in the cache, accuracy moves by at most ±0.3 points in S0, S1 and S2 (sometimes it is even lower than TDA, to check). In S3 the oracle gains +1.61 (RN50) and +0.80 (ViT): in temporally ordered streams, wrong pseudo-labels in the cache do cost accuracy, probably because an error repeats over many consecutive frames.
5. **TDA clearly beats video smoothing:** the best smoothing window adds +3.2 (RN50) and +1.2 (ViT) over CLIP, against about +6 for TDA.

> **Open points.** (1) Per-class cache counts and accuracy over stream windows, to explain the null M2 result and why S2 helps. (2) Why the oracle is sometimes below TDA. (3) S3 is the only setting with room for improvement: a method that keeps temporally correlated errors out of the cache is the natural next step. (4) Paired confidence intervals for the order comparisons. (5) Robustness check on CORe50 350×350, which keeps the session background.

Appendix: hyper-parameter search

8 non-test sessions (1, 2, 4, 5, 6, 8, 9, 11), one frame in five, order S0, 5 seeds, TDA accuracy (%) ± 95% CI.

Config	RN50	ViT-B/16
caltech101 (chosen)	**80.47 ±0.18**	**87.74 ±0.20**
eurosat	80.23 ±0.13	87.20 ±0.18
ucf101	80.04 ±0.10	86.70 ±0.17
dtd	79.84 ±0.12	86.16 ±0.21
sun397	79.84 ±0.12	86.16 ±0.21
oxford_pets	79.79 ±0.11	86.14 ±0.20
fgvc	79.73 ±0.10	85.86 ±0.19
oxford_flowers	79.41 ±0.06	85.32 ±0.18
stanford_cars	79.32 ±0.06	85.24 ±0.17
food101	79.02 ±0.10	84.64 ±0.13
CLIP alone	*77.03*	*80.37*

caltech101 is first with both backbones. With ViT its interval does not overlap with the runner-up (eurosat); with RN50 the two overlap, so they are nearly tied. The choice of config is worth 1.5 points (RN50) and 3.1 points (ViT).